

# **Executive Summary**

## **Nassau Candy-Product Line Profitability & Margin Performance Analysis**

**Prepared For:** Government Stakeholders

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## 1. Project Overview

This study presents a product-line profitability and margin performance analysis of Nassau Candy Distributors using a business analytics approach. The objective of the study is to evaluate product and division-level profitability, identify products exposed to high cost and margin risk, measure profit concentration, assess margin volatility, and translate the findings into actionable management recommendations. The analysis was implemented using Python and Pandas for data preparation and analytical calculations, and Streamlit for interactive dashboard development.

## 2. Overall Portfolio Performance

The analysis of the selected portfolio indicates **\$141,784 in revenue** and **\$93,443 in gross profit**, resulting in an overall **gross margin of 65.9%** and a cost share of sales of **34.1%**. Profit per unit is reported at **\$2.42**, demonstrating strong aggregate profitability. However, profitability is not evenly distributed across the product portfolio, making product-level analysis important for effective management decisions.

## 3. Product-Level Findings

At the product level, substantial variation exists in gross margins and cost exposure. **Everlasting Gobstopper** has the highest displayed gross margin at **80%**, followed by **Hair Toffee at 77.8%** and **Wonka Bar-Nutty Crunch Surprise at 71.3%**. The analysis demonstrates that revenue alone is not sufficient for evaluating product performance because products with high sales volume may generate substantial absolute profit while products with high margins may contribute relatively little total profit.

## 4. Division-Level Findings

Division-level analysis shows that **Chocolate is the dominant division**, generating approximately **\$131,693 in sales and \$88,825 in gross profit with a 67.4% gross margin**. The **Other division generates \$9,663 in sales and \$4,333 in gross profit**, but its gross margin of **44.9%** is substantially lower than the Chocolate and Sugar divisions. This identifies the Other division as an area where cost and pricing strategies should be investigated. The Sugar division records \$427 in sales, \$143 in cost, \$283 in gross profit and a 66.6% gross margin.

## 5. Margin Risk Analysis

The risk analysis classified the **15 reviewed products into 11 Healthy, 3 Cost Risk and 1 High Risk Products**. The most significant concern is **Kazookles**, which has a gross margin of only **7.7%** and a cost-to-sales ratio of **92.3%**. Although the product generates \$93 in gross profit, its low margin and extremely high cost exposure make it the most significant profitability risk in the portfolio. The dashboard therefore recommends a **Discontinuation Review** for Kazookles.

## 6. Cost-Risk Products

The analysis identifies **Fun Dip, Nerds and SweetARTS** as Cost Risk products. These products have comparatively low margins and high cost exposure, resulting in a recommendation to conduct **cost renegotiation**. Management should examine supplier terms, procurement quantities, packaging, freight and other controllable cost drivers to identify opportunities for improving their unit economics.

## 7. Profit Concentration

The Pareto analysis highlights significant profit concentration within the portfolio. Approximately **4 of the 15 products representing 26.7% of the reviewed products, account for approximately 80% of cumulative gross profit**. The leading contributors include Wonka Bar-Scrumdiddlyumptious, Wonka Bar-Triple Dazzle Caramel, Wonka Bar-Milk Chocolate and Wonka Bar-Nutty Crunch Surprise. This concentration indicates that maintaining availability, pricing and supply-chain reliability for high-contribution products is important because disruption to these products could materially affect total profitability.

## 8. Margin Volatility

The dashboard reports **margin volatility of 3.36 percentage points** and shows repeated short-term movements and downward spikes in gross margin. These fluctuations indicate that aggregate profitability may conceal periods of weaker performance. Management should investigate significant margin declines to determine whether they are associated with pricing changes, product mix, cost movements, order-level effects or data-quality issues.

## 9. Management Recommendations

The key recommended actions are to conduct an immediate discontinuation review for Kazookles; renegotiate costs for Fun Dip, Nerds and SweetARTS; protect the leading profit contributors through availability, pricing and supply-chain monitoring; investigate the lower-margin Other division; monitor margin volatility and investigate significant downward movements in gross margin; and maintain the Streamlit dashboard as a recurring management decision-support tool.

## 10. Conclusion

Overall, the study demonstrates that **product profitability should be evaluated using multiple dimensions-including gross margin, cost exposure, absolute profit contribution, concentration and volatility-rather than revenue alone**. The Streamlit dashboard converts transactional data into an interactive decision-support tool that enables management to identify priority products, investigate profitability risks and monitor portfolio performance.

### Key Portfolio Indicators

| KPI                | Result                 |
|--------------------|------------------------|
| Revenue            | \$141,784              |
| Gross Profit       | \$93,443               |
| Gross Margin       | 65.9%                  |
| Cost of Sales      | 34.1%                  |
| Profit per Unit    | \$2.42                 |
| Products Reviewed  | 15                     |
| Healthy Products   | 11                     |
| Cost-Risk Products | 3                      |
| High-Risk Products | 1                      |
| Margin Volatility  | 3.36 percentage points |